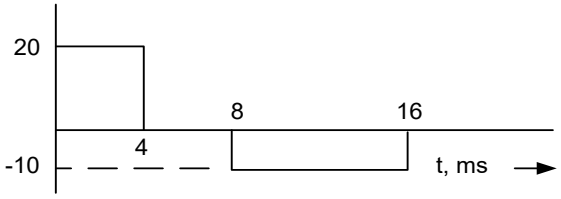
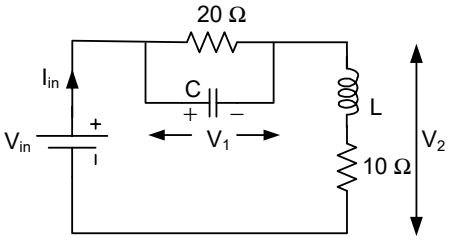
EE1100: Basic Electrical Engineering
Tutorial 2: Transient Analysis of DC Circuits

Q1. (a) Sketch and label the voltage v(t), and instantaneous power p(t) for an 8 μF capacitor whose current waveform in *milli-amps* is plotted in figure below. Assume that $v_{(0)} = 0$.

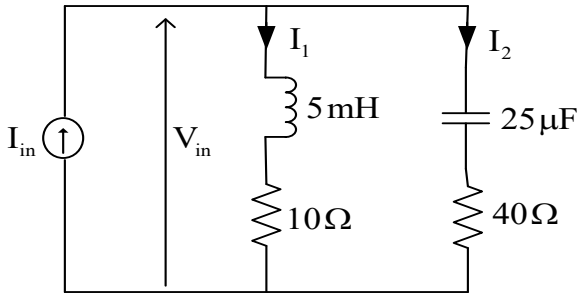


(b) Sketch and label i(t) and instantaneous power p(t) for a 40 mH inductor, whose voltage waveform in *milli-volts* is plotted in Fig. 1. Assume that $i_{(0)} = 0$.

Q2. Find I_{in}, V_1 and V_2 in figure below under DC steady state conditions with V_{in}=15 V. If L=16 mH and the total energy stored in the circuit is 5mJ in the steady state, then find out the value of C.



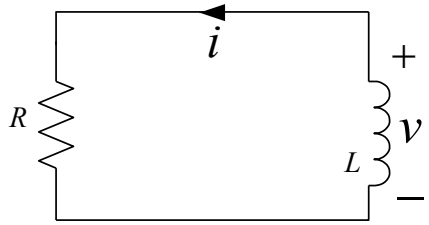
Q3. Find V_{in}, I_1 and I_2 in the figure below, under DC steady state conditions with I_{in} = 2A. Also calculate the total energy stored in the inductor and the capacitor.



Q 4. For the circuit shown here,

$v = 120e^{-50t}$ V

$i = 30e^{-50t}$ for $t > 0$



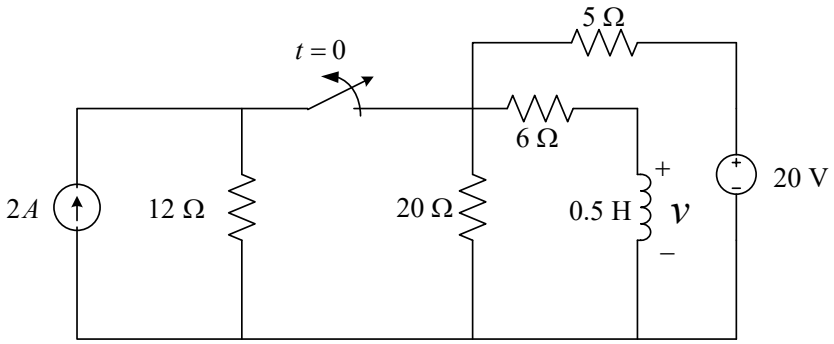
a) Find L and R.

b) Determine the time constant.

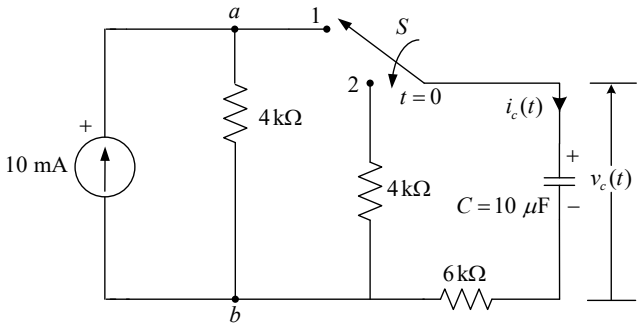
c) Calculate the initial energy in the inductor.

d) Obtain the time it takes to dissipate 50 percent of the initial energy.

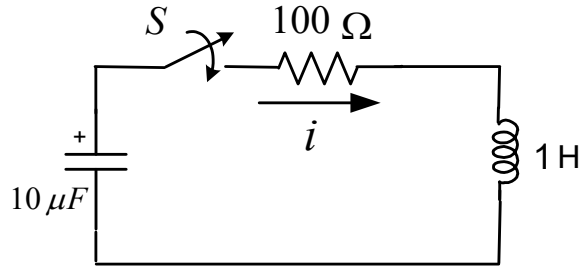
Q5. Find v (t) for t>0 in the circuit given below. (Hint: Current source may be converted into equivalent voltage source to solve the problem).



Q6. The switch S of the circuit, shown in the following figure is thrown from position 1 to 2 at t=0. Find voltage v_c(t) and current i_c(t) expressions for $t \geq 0$.

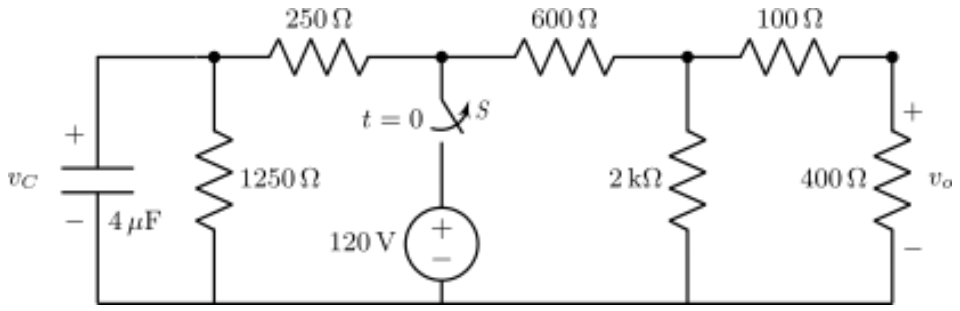


Q. 7. A 5 μF capacitor is discharged suddenly through a coil having an inductance of 2 H and a resistance of 200 Ω. The capacitor is initially charged to 10 V. Compute the following. Find out the following.

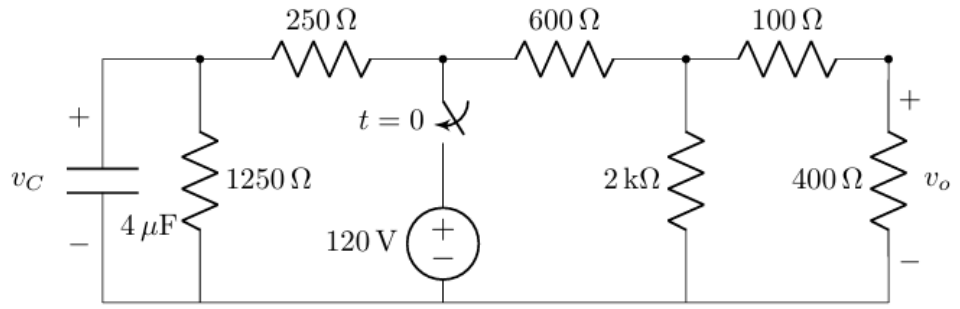


(a) Expression for the current, *i*(t) in the circuit.
(b) Additional resistance required to give critical damping.

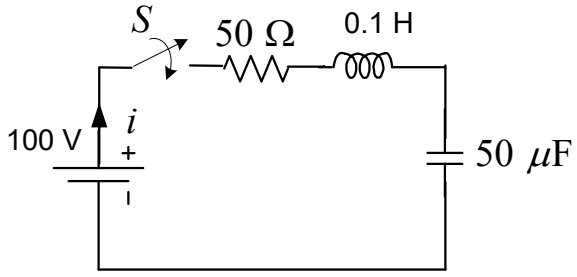
Q8. In the circuit shown below, the switch is opened at t = 0 after having closed for a long time. Determine the capacitor voltage v_c and the output voltage v_o at (a) t = 0−, (b) t = 0+, and (c) t = 1.3 ms.



Q9. In the circuit shown below, the switch is closed at t = 0 after having opened for a long time, as given below. Determine the capacitor voltage v_c and the output voltage v_o at (a) t = 0−, (b) t = 0+, and (c) t = 1.3 ms.



Q10. A series RLC circuit with dc voltage source of 100 V is shown in figure below. Determine critical resistance (R_c), damping factor (ζ), natural frequency (ω_n), damped frequency (ω_d). If the switch is closed at t= 0, then find current, voltage across resistor, inductor and capacitor, assuming zero initial conditions. Indicate whether the circuit is overdamped, critically damped, underdamped or undamped. If resistance element of the circuit is variable, then give its range / value for which circuit response is overdamped, critically damped, underdamped.



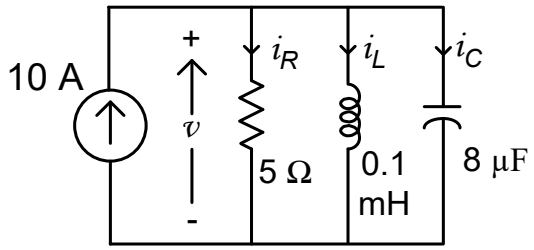
Q11. An engineering device contains two independent energy storage elements subjected to a forcing function, the governing differential equation becomes,

$$2\frac{di}{dt} + 36i + 450\int i\,dt = 900$$

Just prior to the application of the forcing function, the initial value of the response *I* is known to be 10; and the initial value of the integrating device is known to be 70, i.e. $450\int_{-\infty}^{0} i\,dt = 70$.

(a) Draw and label the values of the equivalent circuit for the governing differential equation.
(b) Does the response have a steady-state value different from zero in the time domain?
(c) Find the expression for the complete solution for the current in the time domain.

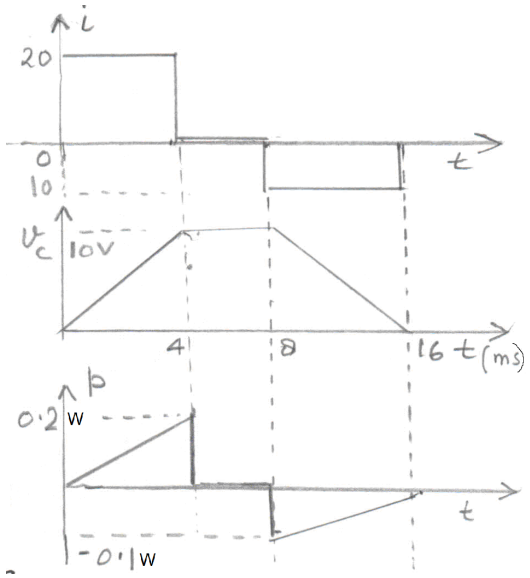
Q12. Consider a three phase RLC parallel circuit as shown below, with R=5Ω, L=0.1mH, and C=8μF.



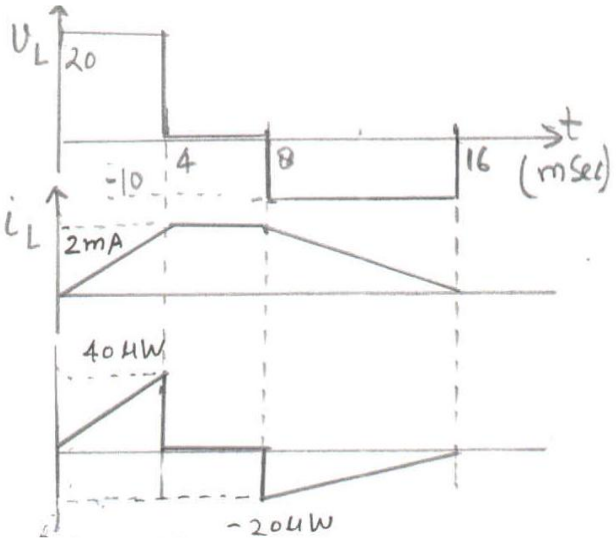
(a) Determine the voltage *v*, across the capacitor.
(b) Compute the currents through all the elements R, L, and C.

Answers: Tutorial 2

1.



(a)



(b)

2. I_{IN}=0.5 A, V_1=10 V, V_2 = 5 V, C= 60 μF
3. V_{in}= 20 V, I_1 = 2 A, I_2=0 A, Energy stored in the inductor = 10 mJ, Energy stored in the capacitor = 5 mJ
4. (a) L= 0.08 H, R= 4 Ω, (b) Time constant = 0.02 s (c) 36 J (d) 6.93 ms
5. $v = v_L = -4\,e^{-20t}$
6. $v_C = 40\,e^{-10\,t}$ V, $\quad i_C = -4\,e^{-10\,t}$ mA
7. i(t) = $\;0.01601\,e^{-50\,t}\sin 312.25\,t\;V$, R_{ext}=1064.9 Ω
8. 100, 38.4 V; 100 V, 25.6 V; 59.45 V, 25.6 V
9. 0 V, 0 V; 0 V, 38.4 V; 70.0 V, 38.4 V
10. $i = 2.7\,e^{-250\,t}\sin 371\,t$ A, $v_R = 135\,e^{-250\,t}\sin 371\,t\;V$, $v_L = 0.1e^{-250\,t}\,[1001.7\cos 371\,t - 675\sin 371\,t\,]\;V$, $v_C = 100 + 0.269e^{-250\,t}\,[-250\sin 371\,t - 371\cos 371\,t]\;V$, R_C=89.44 Ω, $\zeta = 0.559$, $\omega_n = 447.2$ r/s, $\omega_d = 370.8$ r/s, System is underdamped, Critical damping R= 89.44 Ω, Over damping, R > 89.44 Ω, Under damping, 0<R<89.44 Ω, Undamped, R= 0.
11. $i(t) = 28.87\,e^{-9\,t}\sin(12t + 20.27^o)$ A
12. $v(t) = 37.74\,e^{-12500\,t}\sin(33119.1\,t)\;V$, $i_L = 10\left[1 - \dfrac{e^{-12500t}}{0.9367}\sin(33119.10t + 69.51^\circ)\right]$ A

$$i_C = \left(\frac{-I}{\sqrt{1-\zeta^2}}\right)e^{-\zeta\omega_n t}\sin(\omega_d t - \varphi)\text{ A}\quad i_C = \left(\frac{-10}{0.9367}\right)e^{-12500\,t}\sin(33119.10t - 69.51^\circ)\text{ A}$$